A global model of where unrest leads to conflict: model equations

Notation

s, t	0.5° cell, week
$j \in \{\mathrm{C}, \mathrm{U}\}$	outcome: conflict, unrest
$c(s)$, $r(s)$	country and region of cell s (Africa, Middle East, South and Southeast Asia, Latin America)
$y(t)$	calendar year of week t
$g(s)$, $k(t)$	coarse cell (1° or 2°) and time step (Δ weeks) of the space-time fields
$\mathscr{T}_r$	weeks observed in region r (from the start of ACLED coverage to 2025)

Level 1: data

For $t \in \mathscr{T}_{r(s)}$,

$$Y_{jst} \mid \eta_{jst} \sim \text{Bernoulli}\big(\pi_{jst}\big), \qquad \operatorname{logit} \pi_{jst} = \eta_{jst},$$

independent over j, s and t given the linear predictors. $Y_{jst} = 1$ if at least one event of type j is recorded in cell s in week t.

Level 2: linear predictors

$$\begin{aligned}
\eta_{\mathrm{C}st} = {} & \underbrace{x_{st}^\top \beta_{\mathrm{C}} + \alpha_{\mathrm{C},r(s),y(t)}}_{\text{covariates, region-year}} + \underbrace{\delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \delta_{\mathrm{C}}^{nb}\, \bar{h}_{\mathrm{C}st}}_{\text{own history}} + \underbrace{\kappa_{\mathrm{U}\to\mathrm{C}}(s)\, h_{\mathrm{U}st}}_{\text{escalation}} \\
& + \underbrace{u_{\mathrm{C},c(s)} + e_{\mathrm{C},s} + w_{\mathrm{C},s}}_{\text{static}} + \underbrace{v_{\mathrm{C},g(s),k(t)}}_{\text{space-time}}, \\
\eta_{\mathrm{U}st} = {} & x_{st}^\top \beta_{\mathrm{U}} + \alpha_{\mathrm{U},r(s),y(t)} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \delta_{\mathrm{U}}^{nb}\, \bar{h}_{\mathrm{U}st} + \kappa_{\mathrm{C}\to\mathrm{U}}(s)\, h_{\mathrm{C}st} \\
& + u_{\mathrm{U},c(s)} + e_{\mathrm{U},s} + \lambda(s)\, v_{\mathrm{C},g(s),k(t)} + v_{\mathrm{U},g(s),k(t)}.
\end{aligned}$$

Histories (observed, computed from the outcomes; N_{jst} is the number of recorded events):

$$h_{jst} = \log\Big(1 + \textstyle\sum_{t'=t-4}^{t-1} N_{jst'}\Big), \qquad \bar{h}_{jst} = \frac{1}{|R_1(s)|} \sum_{s' \in R_1(s)} h_{js't},$$

with $R_1(s)$ the rook neighbours of s (up to four cells sharing an edge). h_{jst} is the history of the *same* cell over the four weeks before t.

Escalation acts only through the other outcome's history in the same cell over the previous four weeks, with a coefficient that varies in space:

$$\kappa_{\mathrm{U}\to\mathrm{C}}(s) = \kappa_{\mathrm{U}\to\mathrm{C}} + \gamma_{\mathrm{U}\to\mathrm{C}}(s), \qquad \kappa_{\mathrm{C}\to\mathrm{U}}(s) = \kappa_{\mathrm{C}\to\mathrm{U}} + \gamma_{\mathrm{C}\to\mathrm{U}}(s).$$

Terms for the neighbours' unrest, for longer lags (5–26 weeks) and for cells 50–300 km away are left out. The neighbour term is about zero in both fitted regions, and a diagnostic on the fitted latent fields shows no unexplained lagged association between unrest and conflict at 13- and 26-week lags (Supplementary Material).

Level 3: latent effects, given the hyperparameters θ

All terms below are mutually independent, except the two cell effects of the same cell, which are correlated: a place persistently high in conflict tends to be high in unrest as well ($\rho_e > 0$). This replaces a cell effect shared by both outcomes plus one

specific to each: the correlation does the job of the shared part, with the same number of parameters.

country: $u_{j,c} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_{u,j})$,

cell, correlated across outcomes: $\begin{pmatrix} e_{\mathrm{C},s} \\ e_{\mathrm{U},s} \end{pmatrix} \overset{iid}{\sim} \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \sigma^2_{e,\mathrm{C}} & \rho_e \sigma_{e,\mathrm{C}} \sigma_{e,\mathrm{U}} \\ \rho_e \sigma_{e,\mathrm{C}} \sigma_{e,\mathrm{U}} & \sigma^2_{e,\mathrm{U}} \end{pmatrix}\right)$ over cells s,

static field (conflict only): $\boldsymbol{w}_{\mathrm{C}} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{Q}_s(\rho_w, \sigma_w)^{-1}\right)$,

space-time fields: $v_{j,g,k} = \varphi_j \, v_{j,g,k-1} + \sqrt{1-\varphi_j^2}\, \varepsilon_{j,g,k}, \qquad \varepsilon_{j,\cdot,k} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{Q}_s(\rho_j, \sigma_j)^{-1}\right)$,

with $v_{j,g,1} = \varepsilon_{j,g,1}$ (stationary start) and $\varepsilon_{j,\cdot,k}$ independent over k. $\boldsymbol{Q}_s(\rho, \sigma)$ is the finite-volume SPDE precision of a Matérn field ($\nu = 1$, range ρ, SD σ) on the sphere, with sea barriers.

Spatial variation, one of two versions:

	piecewise	smooth
escalation	$\gamma_{\mathrm{U}\to\mathrm{C}}(s) = \gamma_{\mathrm{U}\to\mathrm{C},r(s)}$, fixed effects	$\boldsymbol{\gamma}_{\mathrm{U}\to\mathrm{C}} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{Q}_s(\rho_\gamma, \sigma_\gamma)^{-1}\right)$
loading	$\lambda(s) = \lambda_{r(s)}$	$\lambda(s) = \sum_{m=1}^{M} \vartheta_m \, b_m(s)$, $M \approx 5$–10

Likewise for $\gamma_{\mathrm{C}\to\mathrm{U}}$. The loading $\lambda(s)$ enters the prior precision of the pair $(v_{\mathrm{C}}, \lambda v_{\mathrm{C}})$ (copy construction), whose log-determinant does not depend on it, so the model stays latent Gaussian.

Level 4: priors

parameter	role	prior
β_j, $\alpha_{j,r,y}$, δ_j, δ_j^{nb}, $\kappa_{\mathrm{U}\to\mathrm{C}}$, $\kappa_{\mathrm{C}\to\mathrm{U}}$, γ_r	fixed effects	$\mathcal{N}(0, 10^4)$
$\sigma_{u,j}$, $\sigma_{e,j}$	SDs of iid effects	PC: $P(\sigma > 1) = 0.01$
ρ_e	correlation of the cell effects	$\operatorname{atanh} \rho_e \sim \mathcal{N}(0, 1)$
(ρ_w, σ_w), (ρ_j, σ_j)	Matérn fields	PC: $P(\rho < 100\,\mathrm{km}) = 0.05$, $P(\sigma > 1) = 0.01$
$(\rho_\gamma, \sigma_\gamma)$	varying escalation	PC: $P(\sigma_\gamma > 0.2) = 0.05$
φ_j	persistence	$\operatorname{atanh} \varphi_j \sim \mathcal{N}(0, 1.5^2)$
λ, λ_r or ϑ_m	loadings	$\mathcal{N}(0, 2^2)$

Inference. Empirical Bayes: $\hat{\boldsymbol{\theta}}$ maximises the Laplace approximation of $p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$; the latent effects and fixed effects are summarised by their Gaussian approximation at $\hat{\boldsymbol{\theta}}$; 95% intervals for $\boldsymbol{\theta}$ from the Hessian at $\hat{\boldsymbol{\theta}}$.

Special cases. $\rho_e = 0$, $\lambda(s) \equiv \lambda$, $\gamma \equiv 0$, plus the neighbour cross terms and a static unrest field: the current extended model. With $\rho_e = \lambda = 0$: two separate univariate models.

Static terms, before and after. Conflict: $u_{\mathrm{C},c(s)} + e_{\mathrm{C},s} + w_{\mathrm{C},s}$; unrest: $u_{\mathrm{U},c(s)} + e_{\mathrm{U},s}$. Dropped: the separate shared cell effect (now the correlation ρ_e) and the static unrest field (weak in both regions).

Evidence so far (Africa and the Middle East; West Africa in brackets)

component	estimate, 95% interval	reading
δ_{C}	0.61 [0.59, 0.63] (0.45 [0.41, 0.48])	supported
$\kappa_{\mathrm{U}\to\mathrm{C}}$, no shared cell effect	0.094 [0.065, 0.123] (0.095 [0.032, 0.158])	supported
$\kappa_{\mathrm{U}\to\mathrm{C}}$, with shared cell component	0.057 [0.028, 0.086] (0.061 [−0.001, 0.123])	modest
$\kappa_{\mathrm{U}\to\mathrm{C}}$, shared cell component, $\lambda = 0$	(0.092 [0.029, 0.154])	inflated by 50%
neighbours' unrest (dropped)	−0.011 [−0.021, −0.001] (−0.005 [−0.021, 0.011])	about zero
lagged unrest, 13–26 weeks (dropped)	field diagnostic: corr. −0.04 (−0.08), placebo ≈ 0	no signal
λ (shared episodes)	0.52 [0.46, 0.58] (0.58 [0.45, 0.71]); log-ML vs $\lambda = 0$: (+19.8)	supported
shared cell component	(loading 1.68 [1.28, 2.08], SD 1.02 [0.85, 1.22], range 45 [31, 65] km)	supported; implied $\rho_e \approx 0.92$
$\sigma_{e,\mathrm{C}}$ next to it	(0.16 [0.03, 1.00])	not identified: use ρ_e
φ_{C} per block, range ρ_{C}	0.95 [0.945, 0.963], 453–551 km (0.89 [0.84, 0.93], 426 [356, 510] km)	supported
σ_{U}, φ_{U} (unrest only)	[0.40, 0.50], [0.54, 0.73] (0.55 [0.46, 0.65], 0.70 [0.55, 0.81])	supported
$\sigma_{w,\mathrm{U}}$ (static unrest field)	[0.29, 0.62] ([0.25, 0.99])	weak: dropped
$\gamma(s)$, smooth	semi-synthetic (West Africa): see below	not detectable
$\lambda(s)$	—	not yet tested

Reading of the new evidence (West Africa; Africa and the Middle East pending).

- *Shared episodes* (λ). Setting $\lambda = 0$ lowers the log marginal likelihood by 19.8 and raises $\hat{\kappa}_{\mathrm{U}\to\mathrm{C}}$ from 0.061 to 0.092 (and $\hat{\kappa}_{\mathrm{C}\to\mathrm{U}}$ from 0.076 to 0.110): without the channel, shared episodes are read as escalation. Channel 3 stays.
- *Shared places.* The shared static field has range 45 km, below one cell, so it acts as a shared cell effect, and it leaves $\sigma_{e,\mathrm{C}}$ unidentified (interval 0.03–1.00); the Hessian of the full model at its saved estimate is not positive definite. It is replaced by the correlated cell effects $(e_{\mathrm{C},s}, e_{\mathrm{U},s})$, three parameters for the three identifiable quantities; the saved fit implies $\rho_e \approx 0.92$.
- *Spatially varying escalation.* Outcomes regenerated from the fitted model on the West Africa design, with $\gamma_{\mathrm{U}\to\mathrm{C}}$ a Matérn field. With $\gamma \equiv 0$ (10 replicates) no cell's 95% interval excludes $\hat{\kappa}$ (at most 0.7% of cells), but the posterior-mean spread of $\kappa(s)$ (q90−q10) reaches 0.24. With $\sigma_\gamma = 0.061 = \hat{\kappa}$ (8 replicates; true spread 0.15) the fitted spread is 0.03 (median), at most 1.2% of cells are flagged, and the correlation with the truth is 0.01–0.44. A smooth $\gamma(s)$ of the size of κ is not detectable in one region: we use the piecewise version, one γ_r per region, and read regional differences from intervals, never from maps.